



DATABASE SYSTEMS · AUCA · SPRING 2026

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Hospital Management System

PostgreSQL · PL/pgSQL · Bash · Git

14 Tables · 3NF Normalized · Full Referential Integrity

6 Views · 3 Stored Procedures · Automated Backup

What's Broken in Manual Hospital Systems?



Error-Prone Records

Manual paper records lead to data entry mistakes, lost files, and illegible handwriting



No Room Tracking

Real-time room availability unknown — admissions managed on paper logs



No Centralized History

Patient medical history scattered across departments — no single source of truth



Billing Complexity

Multi-item bills prone to calculation errors, missing charges, audit failures



Scheduling Conflicts

Double-booked appointments and room conflicts without automated checks

A Fully Integrated Database System

14

Normalized Tables

3NF

Normal Form

6

Database Views

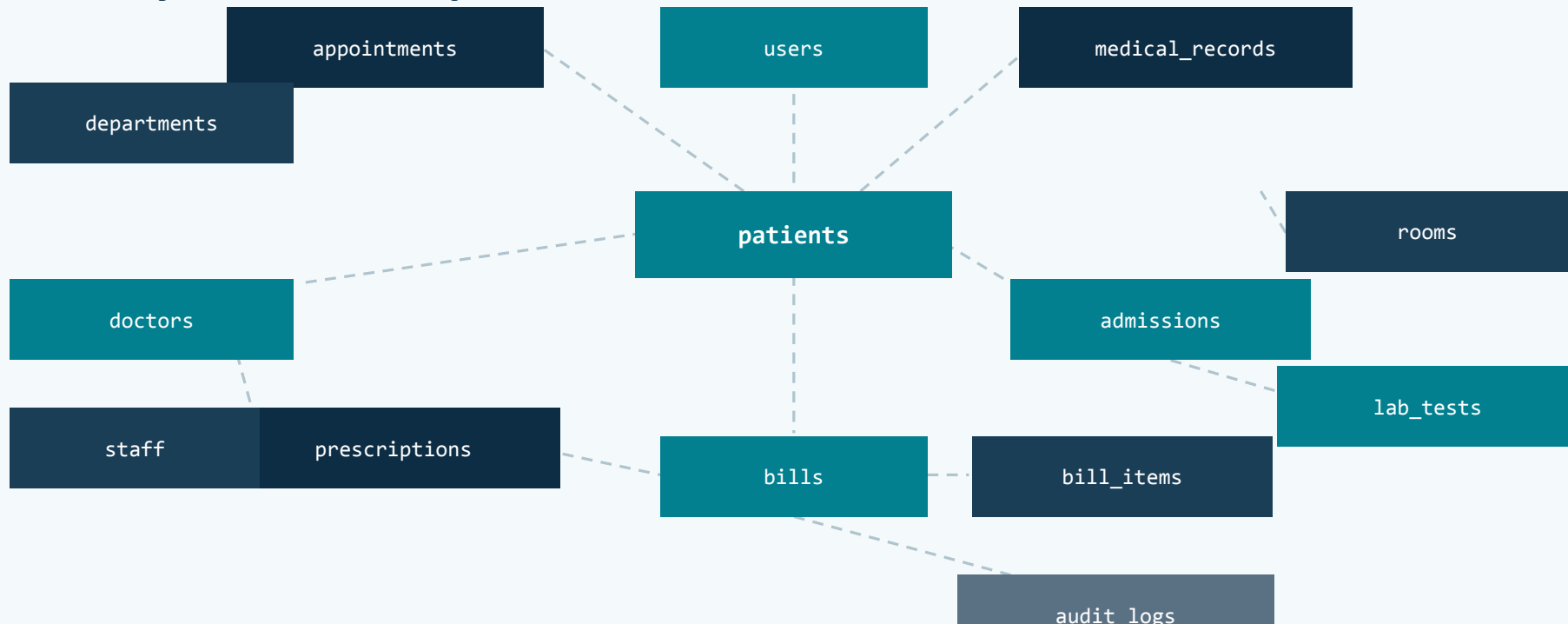
3

Stored Procedures

Key Capabilities

- Patient registration and full medical history tracking
- Appointment scheduling with automated conflict detection
- Hospital admission, room management, and discharge flow
- Prescription, medication, and lab test tracking
- Automated multi-line billing with PL/pgSQL procedures
- Complete audit logging for all data modifications

Entity Relationship Overview



14 tables · All connected via FK constraints · Full referential integrity · 3NF normalized

14-Table Schema — Normalized to 3NF

Core Entities

departments

doctors

patients

staff

Clinical

medical_records

prescriptions

lab_tests

Scheduling

appointments

admissions

rooms

Financial & Audit

bills

bill_items

users

audit_logs

Constraints: CHECK · UNIQUE · NOT NULL · FOREIGN KEY · Indexes: B-tree on FK cols & frequent query columns

Key Technical Decisions



PostgreSQL 16 + PL/pgSQL

- ACID-compliant transactions
- Stored procedures for complex business logic
- PL/pgSQL for conflict detection & billing



3NF Normalization

- Eliminates data redundancy completely
- Ensures referential integrity across tables
- Clean decomposition across all 14 tables



B-tree Indexes

- On all foreign key columns
- On frequent query columns (name, date)
- Balances read performance vs write cost



Audit Logging

- audit_logs table for all data changes
- Timestamps, user ID, old vs new values
- Full traceability for compliance & debugging

PL/pgSQL Stored Procedures

book_appointment()

Conflict Detection

- 1 BEGIN transaction
- 2 Lock doctor's schedule for time slot
- 3 Check for appointment conflicts
- 4 ROLLBACK if conflict found
- 5 INSERT new appointment row
- 6 COMMIT on success

admit_patient()

Room Availability

- 1 BEGIN transaction
- 2 Verify room exists and is vacant
- 3 ROLLBACK if room occupied
- 4 INSERT into admissions table
- 5 UPDATE room status → 'occupied'
- 6 COMMIT all changes atomically

discharge_patient()

Automated Billing

- 1 BEGIN transaction
- 2 Fetch all bill_items for admission
- 3 Calculate total charges auto
- 4 INSERT bills + bill_items rows
- 5 UPDATE room status → 'available'
- 6 UPDATE admission discharge date

All procedures: BEGIN → validate → operate → COMMIT with explicit ROLLBACK on any failure

6 Pre-built Analytical Views

patient_summary

Demographics, contact info, and recent visit history in one queryable view

Cols: name, dob, blood_type, last_visit, doctor

doctor_performance

Appointments per doctor, completion rates, and patient load by department

Cols: doctor_name, dept, total_appts, completed

appointment_summary

Full appointment details joined across patients, doctors, and departments

Cols: patient, doctor, dept, scheduled_at, status

current_admissions

Live view of all admitted patients with room and doctor details

Cols: patient, room_no, admitted, doctor, dept

billing_summary

Bill totals per patient with payment status and outstanding balances

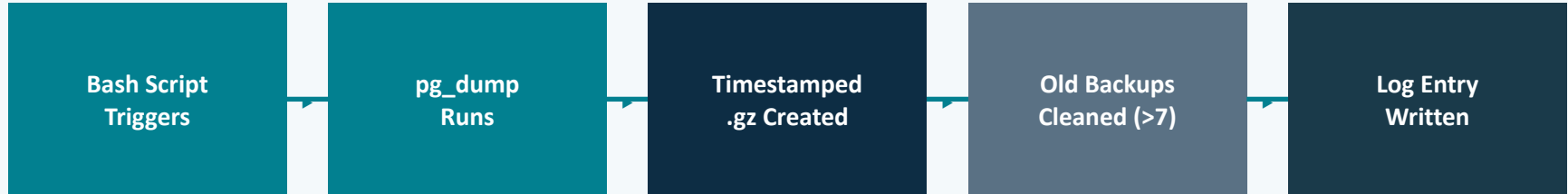
Cols: patient, total_amount, paid, balance, status

lab_test_summary

All lab orders with test type, results, and ordering doctor

Cols: patient, test_type, ordered_by, result, date

Automated Backup & Restore



backup.sh

- Filename: hospital_YYYYMMDD_HHMMSS.sql.gz
- Compressed with gzip — minimal storage
- Retains only last 7 backups — auto-deletes older
- Logs each run with timestamp and file size

restore.sh

- Interactive menu lists all available backups
- User selects target backup file to restore
- Drops & recreates database from the dump file
- Confirmation prompt before destructive action

Technologies Used



PostgreSQL 16

Primary RDBMS

Full ACID compliance, FK enforcement, JSON support, triggers



PL/pgSQL

Procedural Language

Stored procedures, custom functions, exception handling blocks



SQL

Query Language

DML, DDL, complex joins, CTEs, and advanced window functions



Bash Scripting

Automation

Backup scheduling, restore menus, cron-ready log management



Git / GitHub

Version Control

Full commit history, branch management, documented README



macOS

Dev Platform

Local PostgreSQL instance, psql CLI, terminal workflow

github.com/aidaimamvrova/hospital_management

Project Outcomes

14

Tables in 3NF

6

SQL Views

3

Transactions

7

Backup Copies



All 5 hospital problems solved with database-driven solutions



Real-time room availability tracked with atomic transactions



Complete audit trail enables accountability for all data changes



Automatic conflict detection prevents double-booked appointments



Billing errors eliminated via automated PL/pgSQL calculations



Automated backup strategy provides 7 rolling recovery points

A production-ready database design demonstrating real-world hospital management workflows



Thank You

Hospital Management System — Database Systems, AUCA Spring 2026

Student

Aidai Mamyrova

Course

Database Systems, AUCA

GitHub

github.com/aidaimamyrova/hospital_management

Stack

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